

Prediction and Pixel-Space Control from Ridge Regression in Linear Feature Spaces

Theory note, revised with deterministic-equivalent feature-space analysis

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Abstract

We study a noisy linear teacher and a ridge-regression student. The student may be trained after a linear feature map, but accentuation is performed by back-propagating its gradient to the input pixels. The feature map creates two distinct geometries: the feature covariance $C = A\Sigma A^\top$ controls natural-image prediction, whereas $G = AA^\top$ controls pixel-space gradient motion. This distinction gives closed-form finite-sample expressions for generalization, own-path accentuation, and peer review. Weighted one- and two-resolvent deterministic equivalents (DEs) give aggregate prediction and accentuation quantities for an arbitrary fixed linear map. When the map is spectral and aligned with the eigenvectors of Σ , these results further resolve into per-mode weight bias and variance. The same DE risk can also approximate cross-validation and select the ridge penalty. Full PCA, whitening, and top-PC truncation emerge as transparent special cases.

1 Teacher, features, and ridge student

Let the centered input and noisy response be

$$\mathbf{x} \sim \mathcal{N}(0, \Sigma), \quad y = (\boldsymbol{\beta}^\star)^\top \mathbf{x} + \sigma\epsilon, \quad \epsilon \sim \mathcal{N}(0, 1). \quad (1)$$

The noiseless teacher signal variance is

$$S = (\boldsymbol{\beta}^\star)^\top \Sigma \boldsymbol{\beta}^\star. \quad (2)$$

We first map the input to p linear features,

$$\mathbf{z} = A\mathbf{x} \in \mathbb{R}^p, \quad (3)$$

and fit ridge regression in feature coordinates. With the training feature matrix $Z \in \mathbb{R}^{n \times p}$ and $\widehat{C} = Z^\top Z/n$, the normalized-loss convention is

$$\widehat{\boldsymbol{\theta}}_\lambda = \arg \min_{\boldsymbol{\theta}} \left\{ \frac{1}{n} \|\mathbf{y} - Z\boldsymbol{\theta}\|^2 + \lambda \|\boldsymbol{\theta}\|^2 \right\} = (\widehat{C} + \lambda I)^{-1} \frac{Z^\top \mathbf{y}}{n}. \quad (4)$$

The resulting model as a function of pixels is

$$\widehat{f}(\mathbf{x}) = \widehat{\boldsymbol{\theta}}^\top A\mathbf{x} = \widehat{\boldsymbol{\beta}}^\top \mathbf{x}, \quad \widehat{\boldsymbol{\beta}} = A^\top \widehat{\boldsymbol{\theta}}. \quad (5)$$

Thus feature-space estimation and pixel-space attribution are not the same object.

1.1 Best population predictor in the feature span

Define

$$C = A\Sigma A^\top, \quad \mathbf{h} = A\Sigma\boldsymbol{\beta}^*. \quad (6)$$

The best linear feature coefficient and its unrepresented residual are

$$\boldsymbol{\theta}^* = C^\dagger \mathbf{h}, \quad (7)$$

$$r(\mathbf{x}) = (\boldsymbol{\beta}^*)^\top \mathbf{x} - (\boldsymbol{\theta}^*)^\top A\mathbf{x}. \quad (8)$$

By the normal equations, $\text{Cov}(\mathbf{z}, r) = 0$. Since the joint distribution is Gaussian, the residual is independent of the retained features. Its variance is

$$S_\perp = \text{Var}(r) = S - (\boldsymbol{\theta}^*)^\top C \boldsymbol{\theta}^*. \quad (9)$$

Consequently, truncated features reduce the training model to

$$y = (\boldsymbol{\theta}^*)^\top \mathbf{z} + \xi, \quad \text{Var}(\xi) = \sigma_{\text{eff}}^2 = \sigma^2 + S_\perp. \quad (10)$$

The omitted teacher signal is simultaneously irreducible generalization error and additional independent training noise.

It is important not to confuse this prediction-optimal coefficient with the Euclidean projection of the teacher weight. Define

$$G = AA^\top, \quad \mathbf{q} = A\boldsymbol{\beta}^*, \quad (11)$$

and let

$$\boldsymbol{\theta}_E = G^\dagger \mathbf{q} \quad (12)$$

minimize $\|\boldsymbol{\beta}^* - A^\top \boldsymbol{\theta}\|^2$. Then

$$A^\top \boldsymbol{\theta}^* = A^\top (A\Sigma A^\top)^\dagger A\Sigma\boldsymbol{\beta}^*, \quad \text{prediction-optimal}, \quad (13)$$

$$A^\top \boldsymbol{\theta}_E = A^\top (AA^\top)^\dagger A\boldsymbol{\beta}^*, \quad \text{Euclidean-weight-optimal}. \quad (14)$$

The first is the Σ -orthogonal projection onto $\text{range}(A^\top)$; the second is the ordinary Euclidean projection. They agree for isotropic inputs, for an invariant/aligned retained subspace, or when this particular teacher has identical projections, but not in general.

2 Two geometries: prediction versus pixel control

Natural-image prediction is governed by $C = A\Sigma A^\top$. Pixel-space gradient motion is governed by the distinct matrix

$$G = AA^\top, \quad \mathbf{q} = A\boldsymbol{\beta}^*. \quad (15)$$

The vector $\mathbf{h} = A\Sigma\boldsymbol{\beta}^*$ determines population prediction, whereas $\mathbf{q} = A\boldsymbol{\beta}^*$ determines the teacher's response to a back-propagated pixel gradient. Confusing these two vectors hides the prediction-control dissociation.

2.1 Natural-distribution metrics

For a fixed fitted model, the noiseless population prediction error is

$$E_{\text{gen}}(\hat{\boldsymbol{\theta}}) = S_{\perp} + (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*)^{\top} C (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*). \quad (16)$$

Equivalently, if

$$N_{\text{gen}} = \mathbf{h}^{\top} \hat{\boldsymbol{\theta}}, \quad D_{\text{gen}} = \hat{\boldsymbol{\theta}}^{\top} C \hat{\boldsymbol{\theta}}, \quad (17)$$

then

$$\boxed{E_{\text{gen}} = S - 2N_{\text{gen}} + D_{\text{gen}}}. \quad (18)$$

This is an ordinary positive quadratic loss. If performance is measured against a newly observed noisy response rather than the noiseless teacher, one simply adds the irreducible test noise σ^2 . We use the noiseless-teacher coefficient of determination

$$R_{\text{gen}}^2 = 1 - \frac{E_{\text{gen}}}{S}. \quad (19)$$

Its denominator S is a fixed population property. Thus random fitted quantities cannot make the normalization singular. Generalization R^2 can be negative when $E_{\text{gen}} > S$, but it changes continuously with the fitted weight and does not acquire the ratio instability of own-path accentuation. The slope of true response on predicted response over natural inputs is

$$b_{\text{gen}} = \frac{\text{Cov}((\boldsymbol{\beta}^*)^{\top} \mathbf{x}, \hat{\boldsymbol{\theta}}^{\top} A \mathbf{x})}{\text{Var}(\hat{\boldsymbol{\theta}}^{\top} A \mathbf{x})} = \frac{\mathbf{h}^{\top} \hat{\boldsymbol{\theta}}}{\hat{\boldsymbol{\theta}}^{\top} C \hat{\boldsymbol{\theta}}}. \quad (20)$$

2.2 Own-path accentuation metrics

Gradient accentuation from a seed $\mathbf{x}_0$ follows

$$\mathbf{x}(t) = \mathbf{x}_0 + t \nabla_{\mathbf{x}} \hat{f}(\mathbf{x}) = \mathbf{x}_0 + t A^{\top} \hat{\boldsymbol{\theta}}. \quad (21)$$

The student and teacher changes along this path are, respectively,

$$\Delta \hat{f}(t) = t \hat{\boldsymbol{\theta}}^{\top} G \hat{\boldsymbol{\theta}}, \quad (22)$$

$$\Delta f^*(t) = t \mathbf{q}^{\top} \hat{\boldsymbol{\theta}}. \quad (23)$$

Therefore the exact true-on-predicted slope along the path is

$$b_{\text{acc}}(\hat{\boldsymbol{\theta}}) = \frac{\mathbf{q}^{\top} \hat{\boldsymbol{\theta}}}{\hat{\boldsymbol{\theta}}^{\top} G \hat{\boldsymbol{\theta}}}. \quad (24)$$

Suppose the experimenter chooses the distribution of t so that the student's pathwise response variance equals the natural teacher variance S :

$$\text{Var}(t) = \frac{S}{(\hat{\boldsymbol{\theta}}^{\top} G \hat{\boldsymbol{\theta}})^2}. \quad (25)$$

The accentuation calibration error is then

$$\frac{E_{\text{acc}}}{S} = (1 - b_{\text{acc}})^2 = \left(1 - \frac{\mathbf{q}^{\top} \hat{\boldsymbol{\theta}}}{\hat{\boldsymbol{\theta}}^{\top} G \hat{\boldsymbol{\theta}}}\right)^2. \quad (26)$$

When the denominator of R^2 is the variance of the *true* response on the generated path, the exact pathwise statistic becomes

$$R_{\text{acc}}^2 = 1 - \left(1 - \frac{\widehat{\boldsymbol{\theta}}^\top G \widehat{\boldsymbol{\theta}}}{\mathbf{q}^\top \widehat{\boldsymbol{\theta}}}\right)^2. \quad (27)$$

Unlike E_{acc}/S , this statistic is singular when the true path gain $\mathbf{q}^\top \widehat{\boldsymbol{\theta}}$ is close to zero. Hence it can be extremely negative even when other summary metrics appear bounded.

2.3 Two distinct accentuation breakdowns

For clarity, write

$$N_{\text{acc}} = \mathbf{q}^\top \widehat{\boldsymbol{\theta}}, \quad D_{\text{acc}} = \widehat{\boldsymbol{\theta}}^\top G \widehat{\boldsymbol{\theta}} \geq 0. \quad (28)$$

At a fixed, unnormalized path distribution, the raw discrepancy is simply

$$\mathbb{E}_t[(\Delta \widehat{f} - \Delta f^*)^2] = \mathbb{E}[t^2](D_{\text{acc}} - N_{\text{acc}})^2, \quad (29)$$

which is again a regular quadratic quantity. The dramatic behavior enters when the path is normalized by one of its fitted response variances:

$$\frac{E_{\text{acc}}}{S} = \left(1 - \frac{N_{\text{acc}}}{D_{\text{acc}}}\right)^2, \quad \text{singular when } D_{\text{acc}} \rightarrow 0, \quad (30)$$

$$R_{\text{acc}}^2 = 1 - \left(1 - \frac{D_{\text{acc}}}{N_{\text{acc}}}\right)^2, \quad \text{singular when } N_{\text{acc}} \rightarrow 0. \quad (31)$$

These are opposite singular limits. If $N_{\text{acc}} \rightarrow 0$ while D_{acc} remains appreciable, then $R_{\text{acc}}^2 \sim -(D_{\text{acc}}/N_{\text{acc}})^2 \rightarrow -\infty$, whereas $E_{\text{acc}}/S \rightarrow 1$. If $D_{\text{acc}} \rightarrow 0$ while N_{acc} decays more slowly, the student-calibrated loss diverges, whereas $R_{\text{acc}}^2 \rightarrow 0$. Moreover, $N_{\text{acc}} < 0$ means that the teacher moves opposite to the student along the generated path; then $R_{\text{acc}}^2 < 0$ automatically.

There are two mechanisms for approaching these bad regimes. First, finite samples, response noise, and cross-validated regularization can shrink or randomize the signed linear form N_{acc} . At the same time, estimation variance adds a positive contribution to D_{acc} , so the ratio $D_{\text{acc}}/N_{\text{acc}}$ can become very large. Second, a non-aligned dimension-reducing feature map creates a structural mismatch even with infinite noiseless data. In the limit $n \rightarrow \infty$, $\lambda \rightarrow 0$,

$$\widehat{\boldsymbol{\theta}} \longrightarrow \boldsymbol{\theta}^* = C^\dagger \mathbf{h}, \quad (32)$$

and hence

$$N_{\text{acc},\infty} = \mathbf{q}^\top \boldsymbol{\theta}^*, \quad D_{\text{acc},\infty} = (\boldsymbol{\theta}^*)^\top G \boldsymbol{\theta}^*. \quad (33)$$

Because $\mathbf{q} = G\boldsymbol{\theta}_E$ on the effective feature support,

$$N_{\text{acc},\infty} - D_{\text{acc},\infty} = (\boldsymbol{\theta}^*)^\top G(\boldsymbol{\theta}_E - \boldsymbol{\theta}^*). \quad (34)$$

Prediction training selects $\boldsymbol{\theta}^*$, whereas exact pixel-gradient control would prefer $\boldsymbol{\theta}_E$. The two need not agree, and the signed quantity $\mathbf{q}^\top \boldsymbol{\theta}^*$ can be small or even negative despite small natural generalization error.

This structural mechanism is absent in direct pixel regression. With $A = I$,

$$C = \Sigma, \quad G = I, \quad \mathbf{h} = \Sigma \boldsymbol{\beta}^*, \quad \mathbf{q} = \boldsymbol{\beta}^*, \quad S_\perp = 0, \quad \boldsymbol{\theta}^* = \boldsymbol{\theta}_E = \boldsymbol{\beta}^*. \quad (35)$$

Thus infinite-data, zero-ridge pixel regression has $N_{\text{acc}} = D_{\text{acc}} = \|\boldsymbol{\beta}^*\|^2$ and perfect own-path calibration. Its breakdown is entirely a finite-sample/noise/regularization effect, often driven by low-variance input directions. By contrast, a fixed non-aligned truncated feature map can have both that statistical effect and an irreducible projection mismatch. An aligned PCA truncation is an important intermediate case: it has $S_{\perp} > 0$ in general, but its prediction- and Euclidean-optimal weights agree on the retained modes, so its infinite-data own-path calibration remains exact.

3 Deterministic equivalents for an arbitrary fixed feature map

The preceding exact formulas do not require C and G to commute. Their leading aggregate DEs also remain tractable for a deterministic, non-aligned map. Assume p/n has a finite positive limit and the relevant operator norms and deterministic quadratic forms remain bounded. Let κ solve

$$\kappa - \lambda = \frac{\kappa}{n} \text{tr}[C(C + \kappa I)^{-1}], \quad (36)$$

and define

$$\mathcal{R} = (C + \kappa I)^{-1}, \quad H = C\mathcal{R}, \quad \boldsymbol{\mu} = H\boldsymbol{\theta}^*, \quad (37)$$

as well as

$$v_0 = \frac{\kappa^2(\boldsymbol{\theta}^*)^\top \mathcal{R}C\mathcal{R}\boldsymbol{\theta}^* + \sigma_{\text{eff}}^2}{n - \text{tr}(C^2\mathcal{R}^2)}, \quad \sigma_{\text{eff}}^2 = \sigma^2 + S_{\perp}. \quad (38)$$

Then the ordinary generalization quantities have DEs

$$N_{\text{gen}}^{\text{DE}} = \mathbf{h}^\top \boldsymbol{\mu}, \quad (39)$$

$$D_{\text{gen}}^{\text{DE}} = \boldsymbol{\mu}^\top C\boldsymbol{\mu} + v_0 \text{tr}(C^2\mathcal{R}^2), \quad (40)$$

$$\begin{aligned} E_{\text{gen}}^{\text{DE}} &= S - 2N_{\text{gen}}^{\text{DE}} + D_{\text{gen}}^{\text{DE}} \\ &= S_{\perp} + (\boldsymbol{\mu} - \boldsymbol{\theta}^*)^\top C(\boldsymbol{\mu} - \boldsymbol{\theta}^*) + v_0 \text{tr}(C^2\mathcal{R}^2). \end{aligned} \quad (41)$$

The first two terms in the final line are approximation bias and omitted signal; the last is estimation variance.

For pixel accentuation,

$$N_{\text{acc}}^{\text{DE}} = \mathbf{q}^\top \boldsymbol{\mu}, \quad (42)$$

$$D_{\text{acc}}^{\text{DE}} = \boldsymbol{\mu}^\top G\boldsymbol{\mu} + v_0 \text{tr}(G\mathcal{R}C\mathcal{R}). \quad (43)$$

The second line follows from the weighted two-resolvent identity

$$Q_{\lambda} G Q_{\lambda} \asymp \left(\frac{\kappa}{\lambda}\right)^2 \left[\mathcal{R}G\mathcal{R} + \frac{n^{-1} \text{tr}(G\mathcal{R}C\mathcal{R})}{1 - n^{-1} \text{tr}(C^2\mathcal{R}^2)} \mathcal{R}C\mathcal{R} \right], \quad (44)$$

where $Q_{\lambda} = (\hat{C} + \lambda I)^{-1}$. The correction after $\mathcal{R}G\mathcal{R}$ is essential: replacing the two random resolvents independently by their one-point DE misses the random-design contribution. Equations (41) and (43) reduce to the componentwise formulas below when C and G share an eigenbasis.

4 Geometry of protected prediction and catastrophic accentuation

4.1 Exact two-center geometry

The cleanest geometric description uses the two feature-space inner products

$$\langle \mathbf{u}, \mathbf{v} \rangle_C = \mathbf{u}^\top C \mathbf{v}, \quad \langle \mathbf{u}, \mathbf{v} \rangle_G = \mathbf{u}^\top G \mathbf{v}. \quad (45)$$

Natural prediction compares $\hat{\boldsymbol{\theta}}$ with the prediction-optimal center $\boldsymbol{\theta}_P = \boldsymbol{\theta}^* = C^\dagger \mathbf{h}$ in the C metric:

$$E_{\text{gen}} = S_\perp + \|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_P\|_C^2. \quad (46)$$

Pixel control instead compares $\hat{\boldsymbol{\theta}}$ with the Euclidean-weight-optimal center $\boldsymbol{\theta}_E = G^\dagger \mathbf{q}$ in the G metric. On the effective feature support, $\mathbf{q} = G\boldsymbol{\theta}_E$ and therefore

$$N_{\text{acc}} = \langle \hat{\boldsymbol{\theta}}, \boldsymbol{\theta}_E \rangle_G, \quad D_{\text{acc}} = \|\hat{\boldsymbol{\theta}}\|_G^2. \quad (47)$$

Let $B_E = \|\boldsymbol{\theta}_E\|_G^2 > 0$ and make the exact G -orthogonal decomposition

$$\hat{\boldsymbol{\theta}} = a\boldsymbol{\theta}_E + \mathbf{u}, \quad \langle \mathbf{u}, \boldsymbol{\theta}_E \rangle_G = 0. \quad (48)$$

Then

$$N_{\text{acc}} = aB_E, \quad (49)$$

$$D_{\text{acc}} = a^2 B_E + \|\mathbf{u}\|_G^2, \quad (50)$$

$$\frac{D_{\text{acc}}}{N_{\text{acc}}} = a + \frac{\|\mathbf{u}\|_G^2}{aB_E}. \quad (51)$$

Equation (51) separates two control failures. The coefficient a measures signed calibration along the realizable teacher; $\mathbf{u}$ is pixel-gradient energy orthogonal to that teacher. For $a > 0$, $R_{\text{acc}}^2 < 0$ exactly when

$$a + \frac{\|\mathbf{u}\|_G^2}{aB_E} > 2. \quad (52)$$

It becomes arbitrarily negative when $a \rightarrow 0$ with nonzero orthogonal energy, or when the orthogonal energy becomes arbitrarily large relative to aB_E . Natural prediction can nevertheless remain accurate because Eq. (46) measures displacement from a different center, in a different metric.

4.2 Sharp prediction cost of collapsing the true path gain

Assume for the moment that C is positive definite on the effective feature support. The smallest excess prediction loss among all fitted weights whose true accentuation gain vanishes is

$$\begin{aligned} \epsilon_{N=0} &\equiv \min_{\boldsymbol{\theta}: \mathbf{q}^\top \boldsymbol{\theta} = 0} \|\boldsymbol{\theta} - \boldsymbol{\theta}_P\|_C^2 \\ &= \boxed{\frac{(\mathbf{q}^\top \boldsymbol{\theta}_P)^2}{\mathbf{q}^\top C^{-1} \mathbf{q}}}. \end{aligned} \quad (53)$$

The minimizing fitted weight is

$$\boldsymbol{\theta}_{N=0} = \boldsymbol{\theta}_P - \frac{\mathbf{q}^\top \boldsymbol{\theta}_P}{\mathbf{q}^\top C^{-1} \mathbf{q}} C^{-1} \mathbf{q}. \quad (54)$$

Hence a nearly perfect natural predictor can sit arbitrarily close to the R_{acc}^2 singularity when both

$$\frac{S_{\perp}}{S} \ll 1, \quad \frac{\epsilon_{N=0}}{S} \ll 1. \quad (55)$$

The denominator $\mathbf{q}^{\top} C^{-1} \mathbf{q}$ is the susceptibility of the true path gain to a unit of prediction loss. It is large when the control functional $\mathbf{q}$ points strongly into low-variance feature directions. For singular C , the same statement holds after restricting to its support and using the corresponding pseudoinverse, provided the constraint is reachable.

4.3 Generalized low-variance amplification

For any feature-space error $\mathbf{e}$, define the generalized Rayleigh quotient

$$\rho(\mathbf{e}) = \frac{\mathbf{e}^{\top} G \mathbf{e}}{\mathbf{e}^{\top} C \mathbf{e}} = \frac{\|A^{\top} \mathbf{e}\|^2}{(A^{\top} \mathbf{e})^{\top} \Sigma (A^{\top} \mathbf{e})}. \quad (56)$$

The most dangerous direction is measured by

$$\rho_{\max} = \lambda_{\max}(C^{-1/2} G C^{-1/2}), \quad G \preceq \rho_{\max} C. \quad (57)$$

Thus

$$\|\mathbf{e}\|_G^2 \leq \rho_{\max} \|\mathbf{e}\|_C^2. \quad (58)$$

When $\rho_{\max}$ is modest, good prediction controls pixel-gradient error. When it is very large, a direction can be almost invisible to natural risk but enormous in pixel space. A useful sufficient protection diagnostic is

$$\Delta_G = \|\boldsymbol{\theta}_P - \boldsymbol{\theta}_E\|_G + \sqrt{\rho_{\max}(E_{\text{gen}} - S_{\perp})}. \quad (59)$$

If $\Delta_G \ll \|\boldsymbol{\theta}_E\|_G$, then prediction accuracy forces $\hat{\boldsymbol{\theta}}$ into a small G -ball around the control-optimal weight and accentuation is protected. If Δ_G is comparable to or larger than $\|\boldsymbol{\theta}_E\|_G$, natural prediction alone supplies no such guarantee. The first term is structural feature mismatch; the second is statistical error amplified by low natural variance.

The DE has a scale-specific version of the same amplification. Estimation variance contributes

$$V_{\text{gen}} = v_0 \text{tr}(C^2 \mathcal{R}^2), \quad V_{\text{acc}} = v_0 \text{tr}(G \mathcal{R} C \mathcal{R}), \quad (60)$$

so their ratio is

$$\mathcal{A}_{G/C}(\kappa) = \frac{\text{tr}(G \mathcal{R} C \mathcal{R})}{\text{tr}(C^2 \mathcal{R}^2)}. \quad (61)$$

Prediction can remain protected while accentuation variance is large when $V_{\text{gen}} \ll S$ but V_{acc} is comparable to $|N_{\text{acc}}|$; a large $\mathcal{A}_{G/C}(\kappa)$ makes these inequalities compatible.

4.4 CV-selected simplification of the accentuation ratio

Let

$$z_{\text{acc}}^{\text{DE}} = \frac{D_{\text{acc}}^{\text{DE}}}{N_{\text{acc}}^{\text{DE}}} - 1. \quad (62)$$

For an arbitrary fixed map, Eqs. (43) give

$$z_{\text{acc}}^{\text{DE}} = \frac{\boldsymbol{\mu}^{\top} G \boldsymbol{\mu} - \mathbf{q}^{\top} \boldsymbol{\mu} + v_0 T_G}{\mathbf{q}^{\top} \boldsymbol{\mu}}, \quad T_G = \text{tr}(G \mathcal{R} C \mathcal{R}). \quad (63)$$

At an interior minimizer of the DE generalization risk, define

$$B_{2,3}^C = (\boldsymbol{\theta}^\star)^\top C^2 \mathcal{R}^3 \boldsymbol{\theta}^\star, \quad d_{2,3}^C = \text{tr}(C^2 \mathcal{R}^3). \quad (64)$$

The first-order stationarity condition implies

$$v_0 = \kappa \frac{B_{2,3}^C}{d_{2,3}^C}. \quad (65)$$

Using $\mathbf{q} = G\boldsymbol{\theta}_E$, the CV-selected ratio therefore becomes

$$z_{\text{acc,CV}}^{\text{DE}} = \frac{\boldsymbol{\mu}^\top G(\boldsymbol{\mu} - \boldsymbol{\theta}_E)}{\mathbf{q}^\top \boldsymbol{\mu}} + \frac{\kappa T_G}{\mathbf{q}^\top \boldsymbol{\mu}} \frac{B_{2,3}^C}{d_{2,3}^C}. \quad (66)$$

The first term is deterministic shrinkage plus structural projection mismatch; the second is variance inflation at the penalty selected to protect natural prediction. Cross-validation controls the latter only through C and does not optimize the G geometry or remove the first term.

There is a closer analogue of the pixel-regression simplification when C and G commute and $\boldsymbol{\theta}_P = \boldsymbol{\theta}_E = \boldsymbol{\theta}^\star$, as for an aligned spectral feature map. Define

$$B_{a,b}^G = (\boldsymbol{\theta}^\star)^\top G C^a (C + \kappa I)^{-b} \boldsymbol{\theta}^\star, \quad (67)$$

$$d_{a,b}^G = \text{tr}[G C^a (C + \kappa I)^{-b}]. \quad (68)$$

Then

$$N_{\text{acc}}^{\text{DE}} = B_{1,1}^G, \quad D_{\text{acc}}^{\text{DE}} = B_{2,2}^G + v_0 d_{1,2}^G, \quad (69)$$

and the identity $B_{2,2}^G - B_{1,1}^G = -\kappa B_{1,2}^G$ yields

$$z_{\text{acc,CV}}^{\text{DE}} = \kappa \frac{d_{1,2}^G}{B_{1,1}^G} \left(\frac{B_{2,3}^C}{d_{2,3}^C} - \frac{B_{1,2}^G}{d_{1,2}^G} \right). \quad (70)$$

This has the same difference-of-weighted-averages structure as direct pixel regression, but the CV average is selected in the natural prediction geometry C , while the second average is weighted by pixel-control geometry G . Setting $A = I$, so that $C = \Sigma$, $G = I$, and $\boldsymbol{\theta}^\star = \boldsymbol{\beta}^\star$, recovers the pixel-space formula.

Finally, $R_{\text{acc}}^{2,\text{DE}} = 1 - (z_{\text{acc}}^{\text{DE}})^2$ is a typical-fit plug-in, not generally the Monte Carlo mean. The replacement of a random ratio by a ratio of deterministic means requires relative concentration of its denominator. Precisely in the catastrophic regime where N_{acc} approaches or crosses zero, that expansion is nonuniform and rare realizations can dominate the empirical mean. Medians, quantiles, and a joint fluctuation theory for $(N_{\text{acc}}, D_{\text{acc}})$ are then essential.

5 Spectral feature maps and deterministic equivalents

We now specialize to the setting implemented in the accompanying numerical experiments. Write the input eigendecomposition as

$$\Sigma = U \text{diag}(s_1, \dots, s_d) U^\top, \quad b_j = \mathbf{u}_j^\top \boldsymbol{\beta}^\star. \quad (71)$$

For an index set $\mathcal{J}$, let the feature map retain and rescale PCs,

$$z_j = a_j \mathbf{u}_j^\top \mathbf{x}, \quad j \in \mathcal{J}. \quad (72)$$

In the common eigenbasis of prediction and control, define

$$c_j = a_j^2 s_j, \quad g_j = a_j^2, \quad \theta_j^* = \frac{b_j}{a_j}. \quad (73)$$

Then $C = \text{diag}(c_j)$, $G = \text{diag}(g_j)$, and $q_j = g_j \theta_j^*$. The omitted signal and effective noise are

$$S_\perp = \sum_{j \notin \mathcal{J}} s_j b_j^2, \quad \sigma_{\text{eff}}^2 = \sigma^2 + S_\perp. \quad (74)$$

We take $p, n \rightarrow \infty$ with $p/n \rightarrow \gamma$. The effective ridge parameter $\kappa = \kappa(\lambda)$ solves the Silverstein fixed-point equation

$$\kappa - \lambda = \frac{\kappa}{n} \sum_{j \in \mathcal{J}} \frac{c_j}{c_j + \kappa}. \quad (75)$$

Equivalently, the right side is $\gamma \kappa \int c/(c + \kappa) d\nu(c)$ for the empirical feature spectrum ν .

Define

$$h_j = \frac{c_j}{c_j + \kappa}, \quad (76)$$

$$\text{df}_2(\kappa) = \sum_j \frac{c_j^2}{(c_j + \kappa)^2}, \quad (77)$$

$$C_{\text{sig}}(\kappa) = \sum_j \frac{c_j (\theta_j^*)^2}{(c_j + \kappa)^2}, \quad (78)$$

$$v_0(\kappa) = \frac{\kappa^2 C_{\text{sig}}(\kappa) + \sigma_{\text{eff}}^2}{n - \text{df}_2(\kappa)}. \quad (79)$$

The leading componentwise first and second moments of the fitted feature weight are

$$m_j \equiv \mathbb{E}[\hat{\theta}_j] \simeq h_j \theta_j^*, \quad (80)$$

$$v_j \equiv \text{Var}(\hat{\theta}_j) \simeq v_0 \frac{c_j}{(c_j + \kappa)^2}, \quad (81)$$

$$\mathbb{E}[\hat{\theta}_j^2] \simeq m_j^2 + v_j. \quad (82)$$

Thus the per-feature coefficient error is

$$e_j \equiv \mathbb{E}[(\hat{\theta}_j - \theta_j^*)^2] \simeq (1 - h_j)^2 (\theta_j^*)^2 + v_j. \quad (83)$$

The first term is ridge shrinkage bias; the second combines finite-sample signal and response-noise fluctuations.

5.1 DE generalization metrics

Aggregating the componentwise moments gives

$$E_{\text{gen}}^{\text{DE}} = S_{\perp} + \sum_j c_j e_j, \quad (84)$$

$$R_{\text{gen}}^{2,\text{DE}} = 1 - \frac{E_{\text{gen}}^{\text{DE}}}{S}, \quad (85)$$

$$N_{\text{gen}}^{\text{DE}} = \sum_j c_j m_j \theta_j^*, \quad (86)$$

$$D_{\text{gen}}^{\text{DE}} = \sum_j c_j (m_j^2 + v_j), \quad (87)$$

$$b_{\text{gen}}^{\text{DE}} = \frac{N_{\text{gen}}^{\text{DE}}}{D_{\text{gen}}^{\text{DE}}}. \quad (88)$$

5.2 DE pixel-accentuation metrics

The control geometry changes only the aggregation weights:

$$N_{\text{acc}}^{\text{DE}} = \sum_j g_j m_j \theta_j^*, \quad (89)$$

$$D_{\text{acc}}^{\text{DE}} = \sum_j g_j (m_j^2 + v_j), \quad (90)$$

$$b_{\text{acc}}^{\text{DE}} = \frac{N_{\text{acc}}^{\text{DE}}}{D_{\text{acc}}^{\text{DE}}}, \quad (91)$$

$$\frac{E_{\text{acc}}^{\text{DE}}}{S} = (1 - b_{\text{acc}}^{\text{DE}})^2, \quad (92)$$

$$R_{\text{acc}}^{2,\text{DE}} = 1 - \left(1 - \frac{D_{\text{acc}}^{\text{DE}}}{N_{\text{acc}}^{\text{DE}}}\right)^2. \quad (93)$$

Equations (84)–(93) are the main feature-space extension. Natural generalization weights mode j by c_j ; pixel accentuation weights it by g_j . The two agree only in special cases.

The DE above substitutes ratios of deterministic first and second moments for the nonlinear per-fit ratios. It predicts a typical high-dimensional fit but is not, in general, the exact Monte Carlo mean of R_{acc}^2 . This distinction matters whenever N_{acc} can approach zero: rare fits then dominate the empirical mean, while the median can remain close to the leading DE.

5.3 Input-space weight recovery

The recovered pixel coefficient in retained mode j is $\hat{b}_j = a_j \hat{\theta}_j$. Therefore

$$\mathbb{E}[\hat{b}_j] \simeq a_j m_j, \quad (94)$$

$$\text{Var}(\hat{b}_j) \simeq g_j v_j, \quad (95)$$

$$\mathbb{E}\|\hat{\beta} - \beta^*\|^2 \simeq \sum_{j \in \mathcal{J}} g_j e_j + \sum_{j \notin \mathcal{J}} b_j^2. \quad (96)$$

This formula makes the high-frequency mechanism explicit. A low-variance direction can make a negligible contribution $s_j e_j$ to prediction error but a large contribution $g_j e_j$ to the back-propagated

pixel gradient. The DE predicts the mean and variance envelope across modes; it does not predict the signs or texture of one particular random fitted realization.

6 Cross-validated ridge selection from the DE

Many software packages parameterize ridge as

$$\|\mathbf{y} - Z\boldsymbol{\theta}\|^2 + \alpha\|\boldsymbol{\theta}\|^2, \quad (97)$$

so that $\lambda = \alpha/n$ in Eq. (4). For analytic leave-one-out cross-validation, a DE approximation follows the actual training-size change:

1. set $n_{\text{tr}} = n - 1$;
2. for every candidate α , solve Eq. (75) with $\lambda = \alpha/n_{\text{tr}}$ and aspect ratio p/n_{tr} ;
3. compute $E_{\text{gen}}^{\text{DE}}(\alpha; n_{\text{tr}})$ using σ_{eff}^2 ;
4. select $\alpha_{\text{DE-CV}} = \arg \min_{\alpha} E_{\text{gen}}^{\text{DE}}$;
5. refit the final n -sample DE with $\lambda = \alpha_{\text{DE-CV}}/n$.

For K -fold CV, replace $n - 1$ by the fold training size. A dense log-spaced α grid is important: a coarse grid creates artificial jumps in the selected regularization and therefore jagged downstream control metrics. The DE selects ridge for natural predictive risk, just as ordinary CV does; it does not secretly optimize accentuation.

7 PCA, whitening, and top-PC truncation

The spectral formulas expose the behavior of the three feature spaces used in the FFHQ disk-teacher experiment.

Feature map	a_j	c_j	g_j	omitted signal
Full PCA	1	s_j	1	0
Full whitening	$s_j^{-1/2}$	1	s_j^{-1}	0
Top K PCs	$1, j \leq K$	s_j	1	$\sum_{j>K} s_j b_j^2$

Table 1: Prediction and pixel-control quantities for aligned linear features.

Full PCA. An orthogonal change of basis leaves ridge and the pixel-gradient norm unchanged. It is the pixel-space baseline written in interpretable spectral coordinates.

Full whitening. Regression sees an isotropic covariance, $c_j = 1$, but back-propagation weights each mode by $g_j = 1/s_j$. Low-variance, high-frequency PCs can therefore dominate D_{acc} even though they do not dominate natural prediction. The feature representation may look well conditioned while pixel control is ill conditioned.

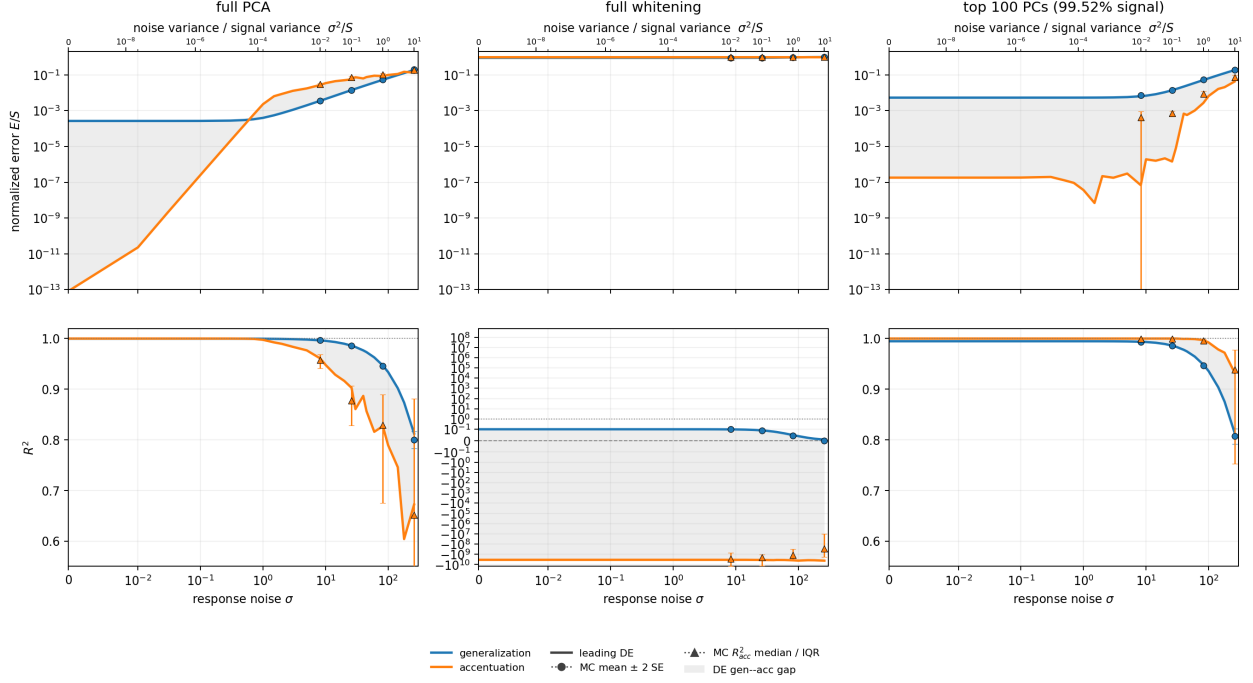


Figure 1: Measured-spectrum FFHQ disk-teacher experiment. Curves are the leading DE; points are Monte Carlo ridge fits. Full PCA develops a generalization–accentuation gap, top-100-PC truncation suppresses the harmful control directions, and full whitening produces a near-zero true path gain and extremely negative R_{acc}^2 . The lower horizontal axis is response-noise standard deviation; the upper axis is σ^2/S .

Top-PC truncation. Discarding low-variance PCs introduces irreducible natural error $S_{\perp}$, but completely prevents those directions from entering the pixel-gradient path. When the teacher signal is concentrated in the leading PCs, a modest predictive cost can buy a large control benefit. The optimal cutoff depends on noise: higher response noise generally favors a smaller, more signal-focused feature space.

8 Peer review across models and feature spaces

Let an evaluator have pixel weight $\hat{\mathbf{w}}_e = A_e^{\top} \hat{\boldsymbol{\theta}}_e$ and a generator have $\hat{\mathbf{w}}_g = A_g^{\top} \hat{\boldsymbol{\theta}}_g$. Along the generator’s path $\mathbf{x}(t) = \mathbf{x}_0 + t\hat{\mathbf{w}}_g$, the teacher and evaluator gains are

$$T_g = (\boldsymbol{\beta}^*)^{\top} \hat{\mathbf{w}}_g, \quad D_{e \leftarrow g} = \hat{\mathbf{w}}_e^{\top} \hat{\mathbf{w}}_g = \hat{\boldsymbol{\theta}}_e^{\top} A_e A_g^{\top} \hat{\boldsymbol{\theta}}_g. \quad (98)$$

If the generator scales its path to its own natural-response target range, the evaluator’s normalized peer error, slope, and R^2 are

$$\frac{E_{e \leftarrow g}}{S} = \left(\frac{D_{e \leftarrow g} - T_g}{\widehat{\mathbf{w}}_g^\top \widehat{\mathbf{w}}_g} \right)^2, \quad (99)$$

$$b_{e \leftarrow g} = \frac{T_g}{D_{e \leftarrow g}}, \quad (100)$$

$$R_{e \leftarrow g}^2 = 1 - \left(1 - \frac{D_{e \leftarrow g}}{T_g} \right)^2. \quad (101)$$

For two independently trained models, a leading peer DE can replace each weight by its deterministic mean in $D_{e \leftarrow g}$ and T_g . Models trained on the same or overlapping samples require a joint two-resolvent DE to retain their correlated fluctuations. Equation (101) also clarifies the observed asymmetry: a generator can move in directions that remain legible to a good evaluator, while the reverse path can have nearly zero teacher gain.

9 Interpretation and limitations

The main conclusions are:

1. Good natural-image prediction does not imply good pixel-space control. Prediction uses C ; control uses G .
2. Low-variance modes are cheap under natural risk but can be expensive under pixel-gradient accentuation, especially after whitening.
3. Top-PC truncation can reverse the generalization–accentuation gap by removing directions that contribute little teacher signal but large control variance.
4. The feature-space DE predicts mean per-mode bias and variance, aggregate natural risk, typical control calibration, and DE-CV ridge selection. It does not specify the visual pattern of a single random fit.
5. R_{acc}^2 and peer R^2 are nonlinear ratios. Near-zero true path gain creates heavy tails, so empirical medians/IQRs and means/standard errors should both be reported.

The aggregate first-order formulas in Section 3 allow a fixed linear map with noncommuting C and G through a weighted two-resolvent DE. The mode-resolved bias and variance formulas in Section 5 additionally assume an aligned spectral map. Full fluctuations of nonlinear ratios, joint peer-review statistics, feature maps learned from the same training sample, and nonlinear features still require joint multi-resolvent or operator-valued deterministic equivalents.

A detailed derivation of the current noise-conditional $\text{Var}(R)$ correction, together with the joint-resolvent covariance calculation needed for the full random-design variance, is given in `notes/accentuation_varR_theory.tex`.